微分学

1. 设 f $[a,b]$上可导，且 $\exists\, c\in(a,b)$，$f'(c)=0$，证：$\exists\, \xi\in(a,b)$，$f'(\xi)=\dfrac{f(\xi)-f(a)}{b-a}$

证：我的想法：令 $g(x)=f'(x)-\dfrac{f(x)-f(a)}{b-a}$ 即证 g $\exists$ 零点 $\xi\in(a,b)$

但缺点是g这个函数生命力不够旺盛，

无g'的性质，且g还不连续 ∴ ✗

正(巧)解：令 $g(x)=\dfrac{f(x)-f(a)}{e^{\frac{x}{b-a}}}$

$$g'(x)=\frac{f'(x)-\frac{f(x)-f(a)}{b-a}}{e^{\frac{x}{b-a}}}$$ 即找 $g'(\xi)=0$

$g'(c)=\dfrac{-1}{b-a}g(c)$ 对g用L中值，$\dfrac{g(c)-g(a)}{c-a}=g'(\eta)$ $(\exists\,\eta\in(a,c))$

$\Rightarrow$ $g'(\eta)=g(c)$ > 异号 ∵ g' s.t. Darboux ∴ $\exists\, g'(\xi)=0$ 证毕

2.(L插值) 设 $a<b<c$，f在R上处处有二阶导

(1) 求二次函数q，使 $q(a)=f(a)$，$q(b)=f(b)$，$q(c)=f(c)$

(2) 证：$\exists\,\xi\in(a,c)$，s.t. $\dfrac{f(a)}{(a-b)(a-c)}+\dfrac{f(b)}{(b-c)(b-a)}+\dfrac{f(c)}{(c-a)(c-b)}=\dfrac{f''(\xi)}{2}$

(3) 证：$\exists\,\eta\in(a,b)$，s.t. $f(a)-2f(\frac{a+b}{2})+f(b)=\dfrac{(b-a)^2}{4}f''(\eta)$

解：(1) $q(x)=\dfrac{(x-b)(x-c)}{(a-b)(a-c)}f(a)+\dfrac{(x-c)(x-a)}{(b-c)(b-a)}f(b)+\dfrac{(x-a)(x-b)}{(c-a)(c-b)}f(c)$

(2) 左 $=\dfrac{g''(x)}{2}$ 令 $g(x)=f(x)-q(x)\xrightarrow{\text{由题}}\exists\,\eta_1\in(a,b)$，$\eta_2\in(b,c)$，$g'(\eta_i)=0$

$\Rightarrow\exists\,\xi\in(\eta_1,\eta_2)$，$g''(\xi)=0$

(3) 法一)

(2)		(3)
a	→	a
b	→	$\frac{a+b}{2}$
c	→	b

(用加权换元化三元关系为二元关系)

法二)

$f(a)=f(\frac{a+b}{2})+f'(\frac{a+b}{2})\frac{a-b}{2}+\frac{1}{2}f''(\xi_1)(\frac{a-b}{2})^2$

$f(b)=f(\frac{a+b}{2})+f'(\frac{a+b}{2})\frac{b-a}{2}+\frac{1}{2}f''(\xi_2)(\frac{a-b}{2})^2$

$\Rightarrow f(a)+f(b)-2f(\frac{a+b}{2})=\dfrac{(a-b)^2}{8}(f''(\xi_1)+f''(\xi_2))$

由Darboux，$\exists\,\eta$

$f''(\eta)=\frac{1}{2}(f''(\xi_1)+f''(\xi_2))$

Date

3. 设 f [a,b] 上处处有一阶导，(a,b) 上处处有二阶导，且 $f'(a)=f'(b)=0$. 证：

$\exists x_0\in(a,b)$, s.t. $|f''(x_0)| \geq \frac{4}{(b-a)^2}|f(b)-f(a)|$　　这样就小了

证：$\cancel{f(x)} = f(a)+f'(a)(x-a)+\frac{1}{2}f''(x_1)(x-a)^2$ ①

$\cancel{f(x)} = f(b)+f'(b)(x-b)+\frac{1}{2}f''(x_2)(x-b)^2$ ②

对f(x)分别在a,b处展开太弱了，$(x-a)^2$, $(x-b)^2$前只有$\frac{1}{2}f''$作系数

①-②, $|f(b)-f(a)| = \frac{1}{2}|f''(x_1)(x-a)^2+f''(x_2)(x-b)^2| \leq \frac{M}{2}\cdot[(x-a)^2+(x-b)^2]$　$= (b-a)^2$

$\leq \cancel{\frac{M}{4}}(b-a)^2$

改：$f(\frac{a+b}{2}) = \begin{cases} f(a)+f'(a)(\quad)+\frac{f''(\xi_1)}{2}\frac{(b-a)^2}{④} & ① \\ f(b)+f'(b)(\quad)+\frac{f''(\xi_2)}{2}\frac{(b-a)^2}{④} & ② \end{cases}$

①-②, $\Rightarrow \frac{4|f(b)-f(a)|}{(b-a)^2} = \frac{1}{2}|f''(\xi_1)-f''(\xi_2)| \leq M$

在端点处取max.

$M=\max\{f''(\xi_1), f''(\xi_2)\}$

而非 $M=\max f''$

∵(a,b)开 ∴不一定∃M

4.(下凸) 设 $f''\geq 0$

证：(1) $\forall x_1<x_2<x_3$, $\frac{f(x_2)-f(x_1)}{x_2-x_1} \leq \frac{f(x_3)-f(x_2)}{x_3-x_2}$

(2) $\forall x<y$, $\alpha\in[0,1]$, $f((1-\alpha)x+\alpha y) \leq (1-\alpha)f(x)+\alpha f(y)$

证：(1) (L中值)　$f(x_2) = f(x_1)+f'(\xi_1)(x_2-x_1)$

$f(x_3) = f(x_2)+f'(\xi_2)(x_3-x_2)$

$\Rightarrow \frac{f(x_2)-f(x_1)}{x_2-x_1} = f'(\xi_1) \leq \underset{\displaystyle \frac{f(x_3)-f(x_2)}{x_3-x_2}}{\underset{\parallel}{f'(\xi_2)}}$

另证：用$f''\geq 0$ 证 $\frac{f(x_2)-f(x_1)}{x_2-x_1} \leq \frac{f(x_3)-f(x_1)}{x_3-x_1}$ ($k_{12}\leq k_{13}$), ⇔ 证 $f(x_2)\leq f(x_1)+\frac{f(x_3)-f(x_1)}{x_3-x_1}(x-x_1)$

记x_2为主元求导即可

(2)　(3)

(2) $x_1 \to x$

$x_2 \to (1-\alpha)x+\alpha y$　$\alpha\in(0,1)$　　证毕

$x_3 \to y$

5. f R上处处有二阶导且 $f(a)=f(b)=0$ 且 $|f''(x)|\leq M$ $\forall x\in[a,b]$

(1) 证：$\forall x\in[a,b]$, $|f(x)| \leq \frac{M}{2}(x-a)(b-x)$

(2) 证：$\forall x\in[a,b]$, $|f(x)| \leq \frac{M}{8}(b-a)^2$

(3) 证：$\forall x\in[a,b]$, $|f'(x)| \leq \frac{M}{2}(b-a)$

∵要证 f(x), f'(x)

对f(a),f(b)在x处展开 右侧留有$f(x_0)$, $f'(x_0)$, $f''(\xi_i)$

证：$0=f(a)= f(x)+f'(x)(a-x)+\frac{1}{2}f''(\xi_1)(a-x)^2$ ①

$0=f(b) = f(x)+f'(x)(b-x)+\frac{1}{2}f''(\xi_2)(b-x)^2$ ②

①·(b-x) - ②·(a-x), (消f')

$(b-a)f(x) = \frac{1}{2}(b-x)(a-x)[f''(\xi_1)(x-a)+f''(\xi_2)(b-x)]$

$\cancel{(b-a)}|f(x)| \leq \frac{(x-a)(b-x)}{2}M\cdot\cancel{(b-a)} \leq \frac{M}{8}(b-a)^2$ ∴(1)(2) ✓

(3) ①-②, $|f'(x)|\cancel{(b-a)} \leq \frac{M}{2}[(a-x)^2+(b-x)^2] \leq \frac{M}{2}(b-a)^2$

KOKUYO

6.(凸. 证不等式)　设 $x_i, y_i, \alpha_i > 0$, $\sum_{i=1}^{n}\alpha_i = 1$

(1) 证: $\ln b \le \ln a + \frac{b-a}{a}$　$(\forall a,b>0)$

(2) 证: $\underline{(\alpha_1 y_1+\cdots+\alpha_n y_n)}\ln\underline{(\alpha_1 y_1+\cdots+\alpha_n y_n)} \le \alpha_1\underline{y_1\ln y_1}+\cdots+\alpha_n\underline{y_n\ln y_n}$

(3) 设 $g(t)=(\alpha_1 x_1^t+\cdots+\alpha_n x_n^t)^{\frac{1}{t}}$, 证 $g(t)\geq 0$

(2) 法一) 令 $\underline{f(x)=x\ln x}$ 下凸　　观察结构啊

$\therefore f(\sum\alpha_i y_i) \le \sum\alpha_i f(y_i)$

法二) $\sum\alpha_i y_i \ln(\underset{b}{\alpha_1 y_1+\cdots+\alpha_n y_n}) \le \sum\alpha_i y_i\left(\ln \underset{a}{y_i} + \frac{\overset{b}{\alpha_1 y_1+\cdots+\alpha_n y_n}\ \overset{-a}{-y_i}}{\underset{a}{y_i}}\right)$

||　　　　||

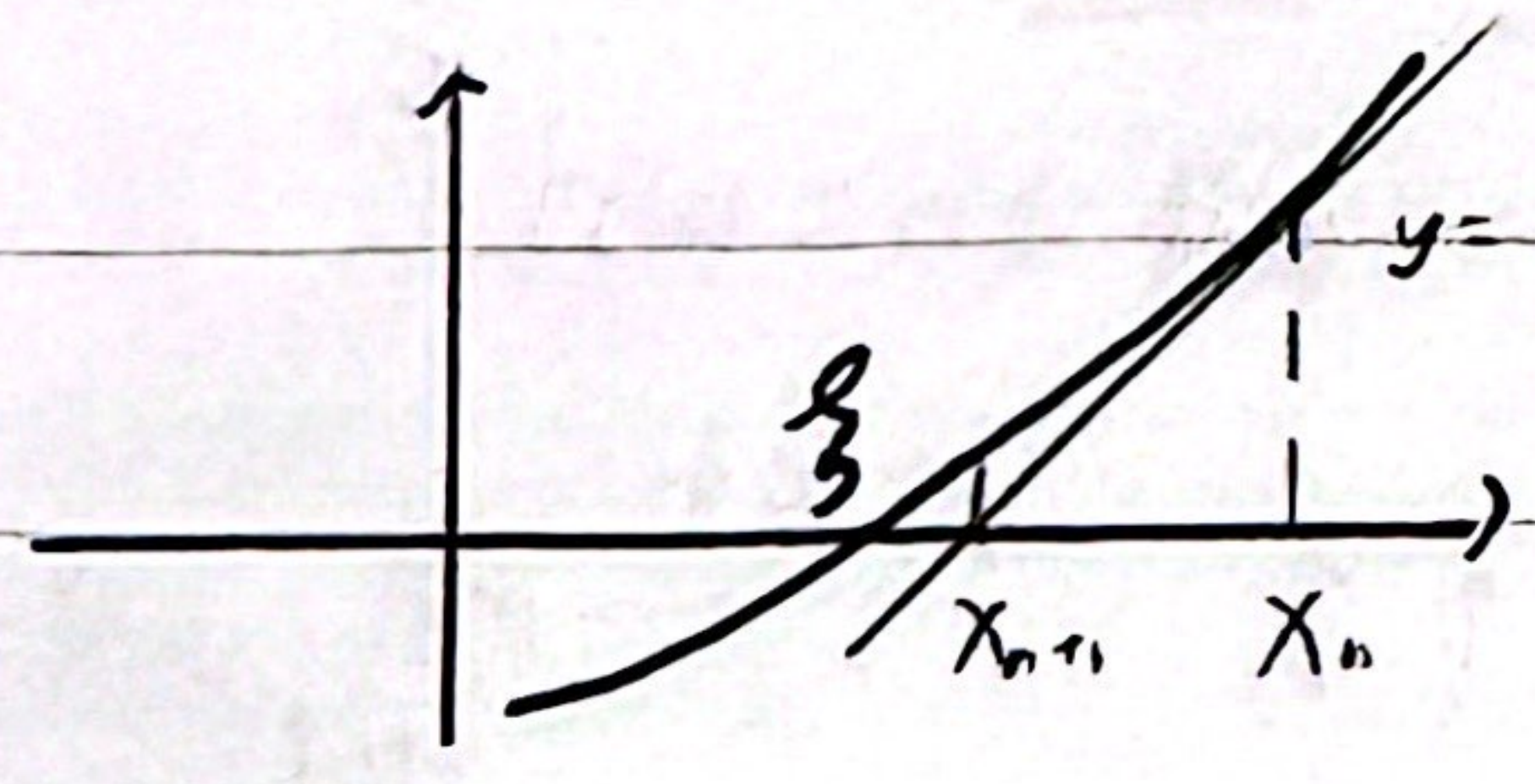
左　　　　右 + $\underbrace{\sum\alpha_i}_{=1}(\sum\alpha_i y_i) - \sum\alpha_i y_i$ = 左

(3) $g(t)=\exp\left(\frac{\ln\sum\alpha_i y_i^t}{t}\right)$

$g'(t)=\exp\left(\frac{\ln\sum\alpha_i y_i^t}{t}\right)\cdot\frac{1}{t^2}\cdot\left(\frac{\sum\alpha_i y_i^t\ln y_i}{\sum\alpha_i y_i^t}t-\ln\sum\alpha_i y_i^t\right)\geq 0$ ✓

7. (Newton切线法) 设 $f\in C^2[a,b]$, f', f''处处>0, $\xi\in(a,b)$ 且 $f(\xi)=0$,

定义 $\{x_n\}_{n=0}^{\infty}$ 为: $x_0=b$, $x_{n+1}=x_n-\frac{f(x_n)}{f'(x_n)}$　$(\forall n\geq 0)$



$y=f'(x_n)(x-x_n)+f(x_n) \xRightarrow{y=0} x_{n+1}=x_n-\frac{f(x_n)}{f'(x_n)}$

(1) 证: $\xi<x_1<b$

(2) 证: $\lim_{n\to\infty}x_n=\xi$

(3) 设 $\min_{a\le x\le b}f'(x)=m$, 证: $|x_n-\xi|\le\frac{f(x_n)}{m}$

(4) 证: $\forall n\in\mathbb{Z}_+$, $\exists c\in(\xi,x_n)$, s.t. $x_{n+1}-\xi=\frac{1}{2}\frac{f''(c)}{f'(x_n)}(x_n-\xi)$

(5) 记 $M=\max_{a\le x\le b}|f''(x)|$, 证: $|x_{n+1}-\xi|\le\frac{M}{2m}(x_n-\xi)^2$

证: (1) $x_1=x_0-\frac{f(x_0)}{f'(x_0)}$　$\because \xi<b=x_0$, $f(\xi)=0$ 且 $f'>0$ $\therefore f(x_0)>f(\xi)=0$

$\Rightarrow x_1<x_0=b$　又$\because$ $\underline{f(x_1)>f(x_0)+f'(x_0)(x_1-x_0)=0=f(\xi)}$ $\therefore x_1>\xi$　$f(\xi$

(2) 假设 $\xi<x_n<x_{n-1}$ $\Rightarrow$ $x_{n+1}=x_n-\frac{f(x_n)}{f'(x_n)}<x_n$　且 $f(x_{n+1})>f'(x_n)+f(x_n)(x_{n+1}-x_n)=$

$\Rightarrow \xi<x_{n+1} \xRightarrow{\text{归纳}} \xi<x_n<x_{n-1}$　$\forall n\in N$ $\xRightarrow{\because\{x_n\}\downarrow\text{有下界}}$ $\lim\exists$(设为l) $\Rightarrow l=l-\frac{f(l)}{f'(l)}$

$\Rightarrow l=\xi$

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(3) 证 $|x_n-\xi| \le \dfrac{f(x_n)}{\min\limits_{[a,b]} f'}$

$\Leftarrow$ 证：$\min\limits_{[a,b]} f' \le \dfrac{f(x_n)-f(\xi)}{x_n-\xi} = f'(\eta)\ (\exists\, \eta\in(\xi,x_n))$ ∴ ✓

(4) $0=f(\xi)=f(x_n)+f'(x_n)(\xi-x_n)+\dfrac{f''(c)}{2}(\xi-x_n)^2$

$\Rightarrow x_n-\xi-\dfrac{f(x_n)}{f'(x_n)}=\dfrac{1}{2}\dfrac{f''(c)}{f'(x_n)}(\xi-x_n)^2$

$x_{n+1}-\xi$ //

(5) 由(4)，✓

8. $f\in C^2$，$\forall x\in\mathbb{R}$，$f(x)\ge 0$

(1) 哪些 $x_0\in\{x \mid \sqrt{f(x)}\ \text{可导}\}$？（提示：$f(x_0)=0$ 的讨论用 Peano）

(2) if $f(0)=f'(0)=0$，$h>0$，设 $M=\max\limits_{[-2h,2h]}|f''|$，用 L 版 Taylor 证：

$\forall x\in[-h,h]$，$f'(x)^2\le 2Mf(x)$

if $f(x_0)\ne 0$ 肯定可导，if $f(x_0)=0$ 就继续讨

(1) $\lim\limits_{x\to x_0}\dfrac{\sqrt{f(x)}-\sqrt{f(x_0)}}{x-x_0}=\lim\limits_{x\to x_0}\dfrac{f(x)-f(x_0)}{x-x_0}\cdot\boxed{\dfrac{1}{\sqrt{f(x)}+\sqrt{f(x_0)}}}=\dfrac{f'(x_0)}{2\sqrt{f(x_0)}}$ （直接求导也行

下面考虑 $f(x_0)=0$ 的情形：claim：$f''(x_0)\ge 0$ if not:

$f(x_0+h)=\underline{f(x_0)}+\underline{f'(x_0)}h+\dfrac{f''(x_0)}{2}h^2+\alpha(h)$ $\quad\lim\limits_{h\to 0}\dfrac{\alpha(h)}{h^2}=0$

∵ $f(x_0)=0\le f(x)$ ∴ x_0 是极小值点 $\Rightarrow f(x_0+h)=\frac{1}{2}f''(x_0)h^2+\alpha(h)$

设 $g(x)=\sqrt{f(x)}$ $\quad\lim\limits_{h\to 0}\dfrac{f(x_0+h)}{h^2}=\dfrac{1}{2}f''(x_0)+\lim\limits_{h\to 0}\dfrac{\alpha(h)}{h^2}=\dfrac{1}{2}f''(x_0)<0\Rightarrow h\to 0,\ f(x_0+h)<0$ ✗

① $f''(x_0)=0$ 时，$g'(x_0)=\lim\limits_{h\to 0}\dfrac{\sqrt{f(x_0+h)}}{h}=\lim\limits_{h\to 0}\left(\operatorname{sgn}(h)\cdot\sqrt{\dfrac{\alpha(h)}{h^2}}\right)=0$ （可导，且 $g'=0$）

② $f''(x_0)>0$ 时，$g'(x_0)=\lim\limits_{h\to 0}\left(\operatorname{sgn}(h)\cdot\sqrt{\frac{1}{2}f''(x_0)}\right)$ $\begin{cases} g'(x_0^+)=\sqrt{\frac{1}{2}f''(x_0)} \\ g'(x_0^-)=-\frac{1}{2}\sqrt{f''(x_0)} \end{cases}$ ∴ 不可导

综上，
$$\begin{cases} f''(x_0)<0 \Rightarrow f(x_0)>0 \Rightarrow g'\ \exists \\ f''(x_0)=0 \Rightarrow g'\ \exists \\ f''(x_0)>0\ (\cancel{\Rightarrow} f(x_0)=0) \Rightarrow g'\ \text{不}\exists \\ \text{且 } f(x_0)=0 \quad \text{推不出} \end{cases}$$

综上：$x\in\{x \mid f''(x_0)>0\ \text{且}\ f(x_0)=0\}$ 时 g' 不可导，else 可导

(2) ① 当 $f(x)=0$ 时 $\xRightarrow{\because f(x)\geq 0}$ $f'(x)=0$ ✓

② $f(x)>0$ 时，$f(x)=f(0)+f'(0)x+\frac{1}{2}f''(c)x^2=\frac{1}{2}f''(c)x^2$

$$\Rightarrow f(x)=\frac{1}{2}f''(c)x^2\leq\frac{M}{2}x^2$$

☆ 取 $y=\sqrt{\frac{2f(x)}{M}}\leq|x|\leq h$

$f(x+y)=f(x)+f'(x)y+\frac{1}{2}f''(\xi_1)y^2\geqslant 0$ 条件 $f\geqslant 0$ 别忘用呀！

$f(x-y)=f(x)+f'(x)(-y)+\frac{1}{2}f''(\xi_2)y^2\geqslant 0$

$$\Rightarrow -\left(\frac{f(x)}{y}+\frac{f''(\xi_1)y}{2}\right)\leq f'(x)\leq\frac{f(x)}{y}+\frac{f''(\xi_2)y}{2}$$

$$\Rightarrow |f'(x)|\leq\max\left\{\frac{f(x)}{y}+\frac{f''(\xi_1)y}{2},\frac{f(x)}{y}+\frac{f''(\xi_2)y}{2}\right\}\leq\frac{f(x)}{y}+\frac{My}{2}=\sqrt{2Mf(x)}$$

我们发现，$0=f(0)=f(x)+f'(x)x+\frac{1}{2}f''(\xi)x^2$

计放缩，只能先把 $f''\to M$
但如此来

$\Rightarrow |f'(x)|=\boxed{\frac{f(x)}{x}+\frac{f''(\xi)x}{2}}$ 这个形式是绕不过去的坎，但它 $\geqslant\sqrt{2Mf(x)}$

∴ 要想出现 $\leq$，就需要先弄出来个 $\sqrt{\frac{2f(x)}{M}}$ （$\frac{f(x)}{y}+\frac{My}{2}$ 的 min 点（关于 y））